8 General Aptitude: Analytical Aptitude (139)

Logic: deduction and induction, Analogy, Numerical relations and reasoning

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximum
1 Mark Count	2	0	0	1	1	1	0	0	0	0.63	2
2 Marks Count	0	2	0	1	1	1	1	1	0	0.88	2
Total Marks	2	4	0	3	3	3	2	2	**0**	**2.38**	**4**

8.1 Age Relation (2)

8.1.1 Age Relation: GATE CSE 2010 | Question: 62

Hari(H), Gita(G), Irfan(I) and Saira(S) are siblings (i.e., brothers and sisters). All were born on 1^{st} January. The age difference between any two successive siblings (that is born one after another) is less than three years. Given the following facts:

i. Hari's age $+$ Gita's age $>$ Irfan's age $+$ Saira's age
ii. The age difference between Gita and Saira is one year. However Gita is not the oldest and Saira is not the youngest.
iii. There are no twins.

In what order they were born (oldest first)?

A. HSIG B. SGHI C. IGSH D. IHSG

gatecse-2010 analytical-aptitude logical-reasoning normal age-relation

Answer key

8.1.2 Age Relation: GATE2013 CE: GA-10

Abhishek is elder to Savar. Savar is younger to Anshul. Which of the given conclusions is logically valid and is inferred from the above statements?

A. Abhishek is elder to Anshul
B. Anshul is elder to Abhishek
C. Abhishek and Anshul are of the same age
D. No conclusion follows

gate2013-ce logical-reasoning age-relation

Answer key

8.2 Code Words (5)

8.2.1 Code Words: GATE CSE 2015 Set 3 | Question: GA-1

If ROAD is written as URDG, then SWANshould be written as:

A. VXDQ B. VZDQ C. VZDP D. UXDQ

gatecse-2015-set3 analytical-aptitude easy logical-reasoning code-words code-language

Answer key

8.2.2 Code Words: GATE CSE 2016 Set 1 | Question: GA04

If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?

A. zentaga B. tagafer. C. tagazen. D. relffer.

gatecse-2016-set1 analytical-aptitude logical-reasoning normal code-words

Answer key

8.2.3 Code Words: GATE2014 EC-4: GA-7

If ‘KCLFTSB’ stands for ‘best of luck’ and ‘SHSWDG’ stands for 'good wishes', which of the following indicates 'ace the exam'?

A. MCHTX B. MXHTC C. XMHCT D. XMHTC

gate2014-ec-4 analytical-aptitude normal logical-reasoning code-words

Answer key☟

8.2.4 Code Words: GATE2018 EE: GA-7

In a certain code, $AMCF$ is written as $EQGJ$ and $NKUF$ is written as $ROYJ$. How will $DHLP$ be written in that code?

A. $RSTN$ B. $TLPH$ C. $HLPT$ D. $XSVR$

gate2018-ee general-aptitude analytical-aptitude logical-reasoning code-words

Answer key☟

8.2.5 Code Words: GATE2019 CE-1: GA-4

If $E=10, J=20, O=30$, and $T=40$, what will be $P+E+S+T$?

A. 51 B. 82 C. 120 D. 164

gate2019-ce-1 analytical-aptitude logical-reasoning code-words

Answer key☟

8.3 Coding Decoding (1)

8.3.1 Coding Decoding: GATE CSE 2025 | Set 2 | GA Question: 7

If IMAGE and FIELD are coded as FHBNJ and EMFJG respectively then, which one among the given options is the most appropriate code for BEACH?

A. CEADP B. IDBFC C. JGIBC D. IBCEC

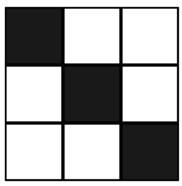
gatecse2025-set2 analytical-aptitude coding-decoding two-marks

Answer key☟

8.4 Counting Figure (1)

8.4.1 Counting Figure: GATE Civil 2023 Set 2 | GA Question: 3

In how many ways can cells in a 3×3 grid be shaded, such that each row and each column have exactly one shaded cell? An example of one valid shading is shown.



A. 2 B. 9 C. 3 D. 6

gatecivil-2023-set2 analytical-aptitude counting-figure

Answer key☟

8.5 Direction Sense (12)

8.5.1 Direction Sense: GATE CSE 2015 Set 2 | Question: GA-7

Four branches of a company are located at $\mathrm{M, N, O}$ and P. M is north of N at a distance of $4\ \mathrm{km}$; P is south of O at a distance of $2\ \mathrm{km}$; N is southeast of O by $1\ \mathrm{km}$. What is the distance between M and P in km?

A. 5.34 B. 6.74 C. 28.5 D. 45.49

gatecse-2015-set2 analytical-aptitude normal direction-sense

Answer key☟

8.5.2 Direction Sense: GATE CSE 2017 Set 2 | Question: GA-3

There are five buildings called V, W, X, Y and Z in a row (not necessarily in that order). V is to the West of W. Z is to the East of X and the West of V. W is to the West of Y. Which is the building in the middle?

A. V B. W C. X D. Y

gatecse-2017-set2 analytical-aptitude direction-sense normal

Answer key

8.5.3 Direction Sense: GATE CSE 2019 | Question: GA-10

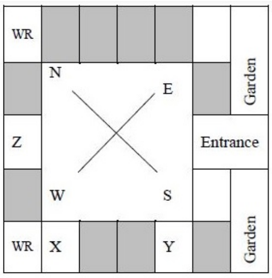
Three of the five students are allocated to a hostel put in special requests to the warden, Given the floor plan of the vacant rooms, select the allocation plan that will accommodate all their requests.

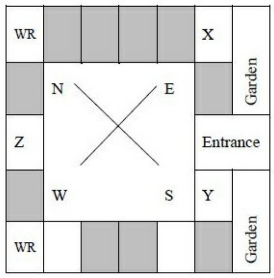
Request by X: Due to pollen allergy, I want to avoid a wing next to the garden.

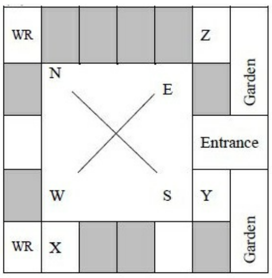
Request by Y: I want to live as far from the washrooms as possible, since I am very much sensitive to smell.

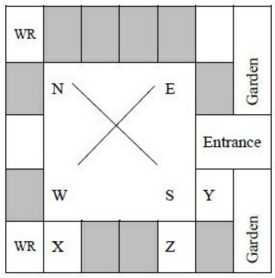
Request by Z: I believe in Vaastu and so I want to stay in South-West wing.

The shaded rooms are already occupied. WR is washroom

A. 

B. 

C. 

D. 

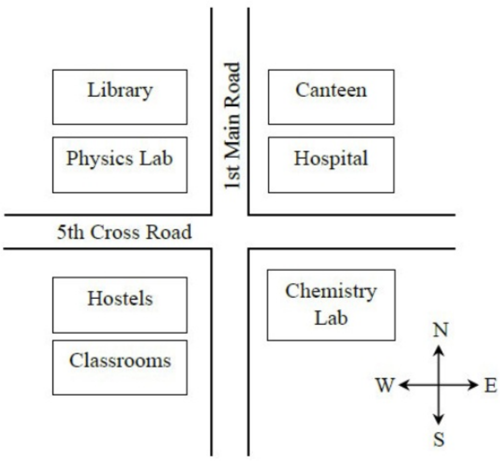
gatecse-2019 general-aptitude analytical-aptitude direction-sense two-marks

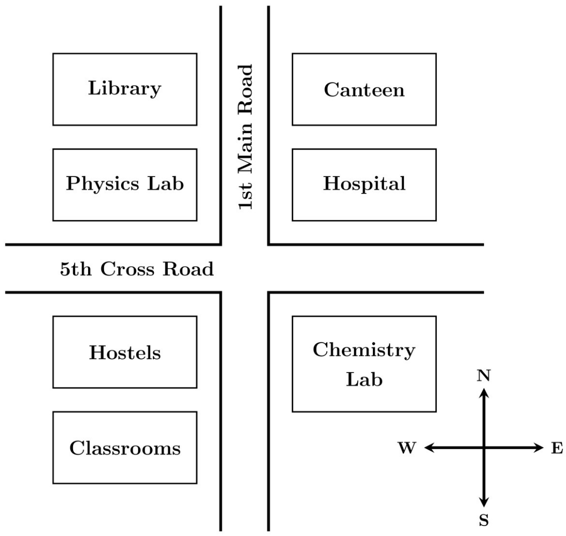
Answer key

8.5.4 Direction Sense: GATE CSE 2025 | Set 1 | GA Question: 5

According to the map shown in the figure, which one of the following statements is correct?

Note: The figure shown is representative.





A. The library is located to the northwest of the canteen.
B. The hospital is located to the east of the chemistry lab.
C. The chemistry lab is to the southeast of physics lab.
D. The classrooms and canteen are next to each other.

gatecse2025-set1 analytical-aptitude direction-sense one-mark

Answer key☟

8.5.5 Direction Sense: GATE Civil 2021 Set 2 | GA Question: 6

On a planar field, you travelled 3 units East from a point O. Next you travelled 4 units South to arrive at point P. Then you travelled from P in the North-East direction such that you arrive at a point that is 6 units East of point O. Next, you travelled in the North-West direction, so that you arrive at point Q that is 8 units North of point P.
The distance of point Q to point O, in the same units, should be _______________

A. 3 B. 4 C. 5 D. 6

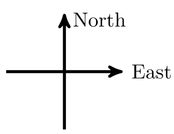
gatecivil-2021-set2 analytical-aptitude direction-sense

Answer key☟

8.5.6 Direction Sense: GATE Civil 2022 Set 2 | GA Question: 10

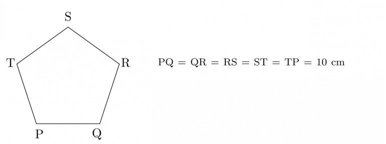
An ant walks in a straight line on a plane leaving behind a trace of its movement. The initial position of the ant is at point P facing east.

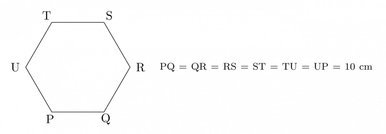
The ant first turns 72° anticlockwise at P, and then does the following two steps in sequence exactly FIVE times before halting.

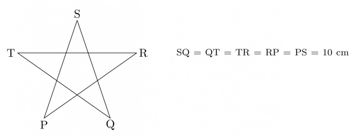


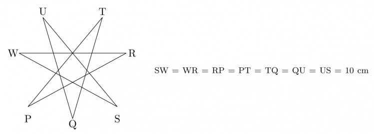
1. moves forward for 10 cm.
2. turns 144° clockwise.

The pattern made by the trace left behind by the ant is

A. 

B. 

C. 

D. 

gatecivil-2022-set2 analytical-aptitude direction-sense

Answer key☟

8.5.7 Direction Sense: GATE Mechanical 2021 Set 1 | GA Question: 2

Ms. X came out of a building through its front door to find her shadow due to the morning sun failing to her right side with the building to her back. From this, it can be inferred that building is facing ________

A. North B. East C. West D. South

gateme-2021-set1 analytical-aptitude direction-sense

Answer key☟

8.5.8 Direction Sense: GATE Mechanical 2021 Set 2 | GA Question: 5

The front door of $\mathrm{Mr.\ X}$'s house faces East. $\mathrm{Mr.\ X}$ leaves the house, walking $50\,\mathrm{m}$ straight from the back door that is situated directly opposite to the front door. He then turns to his right, walks for another $50\,\mathrm{m}$ and stops. The direction of the point $\mathrm{Mr.\ X}$ is now located at with respect to the starting point is ____

A. South-East B. North-East C. West D. North-West

gateme-2021-set2 analytical-aptitude direction-sense

Answer key☟

8.5.9 Direction Sense: GATE2014 AG: GA-9

X is 1 km northeast of Y. Y is 1 km southeast of Z. W is 1 km west of Z. P is 1 km south of W. Q is 1 km east of P. What is the distance between X and Q in km?

A. 1 B. $\sqrt{2}$ C. $\sqrt{3}$ D. 2

gate2014-ag analytical-aptitude direction-sense normal

Answer key☟

8.5.10 Direction Sense: GATE2015 CE-2: GA-4

Mr. Vivek walks 6 meters North-east, then turns and walks 6 meters South-east, both at 60 degrees to east. He further moves 2 meters South and 4 meters West. What is the straight distance in meters between the point he started from and the point he finally reached?

A. $2\sqrt{2}$ B. 2 C. $\sqrt{2}$ D. $1/\sqrt{2}$

gate2015-ce-2 analytical-aptitude general-aptitude direction-sense

Answer key☟

8.5.11 Direction Sense: GATE2016 EC-2: GA-9

M and N start from the same location. M travels 10 km East and then 10 km North-East. N travels 5 km South and then 4 km South-East. What is the shortest distance (in km) between M and N at the end of their travel?

A. 18.60 B. 22.50 C. 20.61 D. 25.00

gate2016-ec-2 direction-sense analytical-aptitude

Answer key☟

8.5.12 Direction Sense: GATE2017 EC-2: GA-4

Fatima starts from point P, goes North for 3 km, and then East for 4 km to reach point Q. She then turns to face point P and goes 15 km in that direction. She then goes North for 6 km. How far is she from point P, and in which direction should she go to reach point P?

A. 8 km, East B. 12 km, North C. 6k m, East D. 10 km, North

gate2017-ec-2 general-aptitude analytical-aptitude direction-sense

Answer key☟

8.6 Family Relationship (7)

8.6.1 Family Relationship: GATE CSE 2025 | Set 1 | GA Question: 4

Consider the relationships among $\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}$, and T :

- P is the brother of Q.
- S is the daughter of Q.
- T is the sister of S.
- R is the mother of Q.

The following statements are made based on the relationships given above.

1. R is the grandmother of S.
2. P is the uncle of S and T.
3. R has only one son.
4. Q has only one daughter.

Which one of the following options is correct?

A. Both (1) and (2) are true.
B. Both (1) and (3) are true.
C. Only (3) is true.
D. Only (4) is true.

gatecse2025-set1 analytical-aptitude logical-reasoning family-relationship one-mark

Answer key☟

8.6.2 Family Relationship: GATE Civil 2021 Set 1 | GA Question: 6

Statement: Either P marries Q or X marries Y

Among the options below, the logical NEGATION of the above statement is :

A. P does not marry Q and X marries Y
B. Neither P marries Q nor X marries Y
C. X does not marry Y and P marries Q
D. P marries Q and X marries Y

gatecivil-2021-set1 analytical-aptitude logical-reasoning family-relationship

Answer key☟

8.6.3 Family Relationship: GATE Civil 2022 Set 1 | GA Question: 4

Given the statements:

- P is the sister of Q.
- Q is the husband of R.
- R is the mother of S.
- T is the husband of P.

Based on the above information, T is ____________ of S.

A. the grandfather B. an uncle C. the father D. a brother

gatecivil-2022-set1 analytical-aptitude logical-reasoning family-relationship

Answer key☟

8.6.4 Family Relationship: GATE2016 EC-3: GA-3

M has a son **Q** and a daughter **R**. He has no other children. **E** is the mother of **P** and daughter-inlaw of **M**. How is **P** related to **M**?

A. **P** is the son-in-law of **M**.
B. **P** is the grandchild of **M**.
C. **P** is the daughter-in law of **M**.
D. **P** is the grandfather of **M**.

gate2016-ec-3 family-relationship logical-reasoning

Answer key☟

8.6.5 Family Relationship: GATE2017 EC-2: GA-7

Each of $P, Q, R, S, W, X, Y$and Z has been married at most once. X and Y are married and have two children P and Q. Z is the grandfather of the daughter S of P. Further, Z and W are married and are parents of R. Which one of the following must necessarily be FALSE?

A. X is the mother-in-law of R
B. P and R are not married to each other
C. P is a son of X and Y
D. Q cannot be married to R

gate2017-ec-2 general-aptitude logical-reasoning family-relationship

Answer key☟

8.6.6 Family Relationship: GATE2019 CE-1: GA-10

P, Q, R, S, and T are related and belong to the same family. P is the brother of S, Q is the wife of P. R and T are the children of the siblings P and S respectively. Which one of the following statement is necessarily FALSE?

A. S is the aunt of R
B. S is the aunt of T
C. S is the sister-in-law of Q
D. S is the brother of P

gate2019-ce-1 general-aptitude logical-reasoning family-relationship

Answer key☟

8.6.7 Family Relationship: GATE2019 ME-1: GA-10

M and N had four children P, Q, R and S. Of them, only P and R were married. They had children X and Y respectively. If Y is a legitimate child of W, which of the following statements is necessarily FALSE?

A. M is the grandmother of Y
B. R is the father of Y
C. W is the wife of R
D. W is the wife of P

gate2019-me-1 analytical-aptitude family-relationship

Answer key☟

8.7 Inequality (1)

8.7.1 Inequality: GATE2014 EC-3: GA-5

In which of the following options will the expression $P < M$ be definitely true?

A. $M < R > P > S$
B. $M > S < P < F$
C. $Q < M < F = P$
D. $P = A < R < M$

gate2014-ec-3 logical-reasoning analytical-aptitude inequality

Answer key☟

8.8 Logical Inference (1)

8.8.1 Logical Inference: GATE CSE 2025 | Set 2 | GA Question: 6

Based only on the conversation below, identify the logically correct inference:
"Even if I had known that you were in the hospital, I would not have gone there to see you", Ramya told Josephine.

A. Ramya knew that Josephine was in the hospital.
B. Ramya did not know that Josephine was in the hospital.

C. Ramya and Josephine were once close friends; but now, they are not.
D. Josephine was in the hospital due to an injury to her leg.

gatecse2025-set2 analytical-aptitude logical-inference two-marks

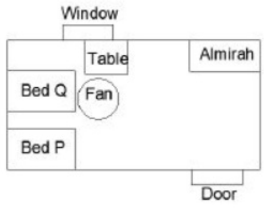
Answer key

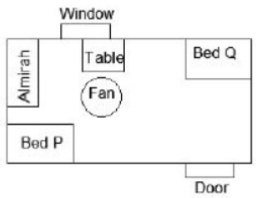
8.9 Logical Reasoning (39)

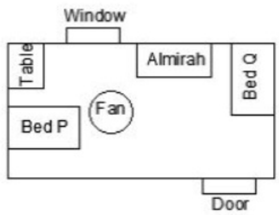
8.9.1 Logical Reasoning: GATE 2024 Electrical | GA Question: 2

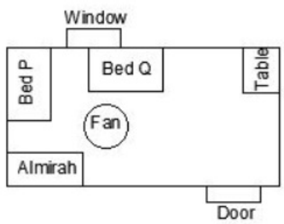
P and Q have been allotted a hostel room with two beds, a study table, and an almirah. P is an avid bird-watcher and wants to sit at the table and watch birds outside the window. Q does not mind that as long as his bed is close to the ceiling fan.

Which one of the following arrangements suits them the most?

A. 

B. 

C. 

D. 

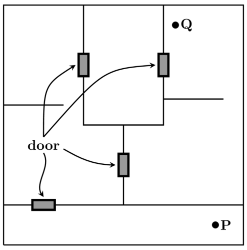
gate2024-ee analytical-aptitude logical-reasoning

Answer key

8.9.2 Logical Reasoning: GATE CH 2022 | GA Question: 5

A building has several rooms and doors as shown in the top view of the building given below. The doors are closed initially.

What is the minimum number of doors that need to be opened in order to go from the point P to the point Q?



A. 4 B. 3 C. 2 D. 1

gatech-2022 analytical-aptitude logical-reasoning

Answer key

8.9.3 Logical Reasoning: GATE CSE 2010 | Question: 61

If $137+276=435$ how much is $731+672$?

A. 534 B. 1403 C. 1623 D. 1513

gatecse-2010 analytical-aptitude normal logical-reasoning

Answer key

8.9.4 Logical Reasoning: GATE CSE 2015 Set 3 | Question: GA-7

The head of newly formed government desires to appoint five of the six selected members P, Q, R, S, T and U to portfolios of Home, Power, Defense, Telecom, and Finance. U does not want any portfolio if S gets one of the five. R wants either Home or Finance or no portfolio. Q says that if S gets Power or Telecom, then she must get the other one. T insists on a portfolio if P gets one.

Which is the valid distribution of portfolios?

A. P-Home, Q-Power, R-Defense, S-Telecom, T-Finance
B. R-Home, S-Power, P-Defense, Q-Telecom, T-Finance
C. P-Home, Q-Power, T-Defense, S-Telecom, U-Finance
D. Q-Home, U-Power, T-Defense, R-Telecom, P-Finance

gatecse-2015-set3 analytical-aptitude normal logical-reasoning

Answer key

8.9.5 Logical Reasoning: GATE CSE 2017 Set 2 | Question: GA-7

There are three boxes. One contains apples, another contains oranges and the last one contains both apples and oranges. All three are known to be incorrectly labeled. If you are permitted to open just one box and then pull out and inspect only one fruit, which box would you open to determine the contents of all three boxes?

A. The box labeled 'Apples'
B. The box labeled 'Apples and Oranges'
C. The box labeled 'Oranges'
D. Cannot be determined

gatecse-2017-set2 analytical-aptitude normal tricky logical-reasoning

Answer key

8.9.6 Logical Reasoning: GATE CSE 2021 Set 2 | GA Question: 10

Six students $\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}, \mathrm{T}$ and U, with distinct heights, compare their heights and make the following observations.

- Observation I: S is taller than R.
- Observation II: Q is the shortest of all.
- Observation III: U is taller than only one student.
- Observation IV: T is taller than S but is not the tallest

The number of students that are taller than R is the same as the number of students shorter than ____________.

A. T B. R C. S D. P

gatecse-2021-set2 analytical-aptitude logical-reasoning two-marks

Answer key☟

8.9.7 Logical Reasoning: GATE CSE 2023 | GA Question: 4

A survey for a certain year found that 90% of pregnant women received medical care at least once before giving birth. Of these women, 60% received medical care from doctors, while 40% received medical care from other healthcare providers.

Given this information, which one of the following statements can be inferred with *certainty*?

A. More than half of the pregnant women received medical care at least once from a doctor.
B. Less than half of the pregnant women received medical care at least once from a doctor.
C. More than half of the pregnant women received medical care at most once from a doctor.
D. Less than half of the pregnant women received medical care at most once from a doctor.

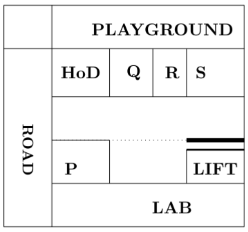
gatecse-2023 analytical-aptitude logical-reasoning one-mark

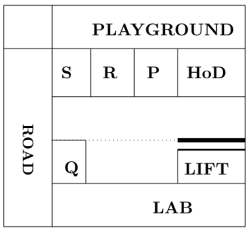
Answer key☟

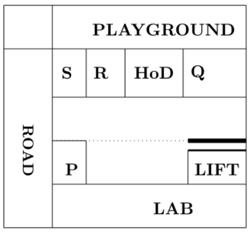
8.9.8 Logical Reasoning: GATE Civil 2020 Set 2 | GA Question: 4

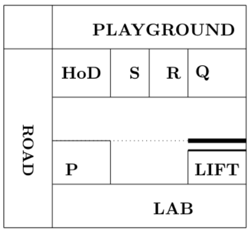
After the inauguration of the new building, the Head of the Department (HoD) collated faculty preferences for office space. P wanted a room adjacent to the lab. Q wanted to be close to the lift. R wanted a view of the playground and S wanted a corner office.

Assuming that everyone was satisfied, which among the following shows a possible allocation?

A. 

B. 

C. 

D. 

gate2020-ce-2 analytical-aptitude logical-reasoning

Answer key☟

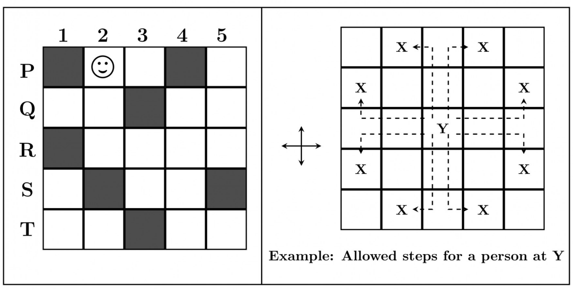
8.9.9 Logical Reasoning: GATE Civil 2022 Set 1 | GA Question: 9

In the square grid shown on the left, a person standing at $\mathrm{P2}$ position is required to move to $\mathrm{P5}$ position.

The only movement allowed for a step involves, "two moves along one direction followed by one move in a perpendicular direction". The permissible directions for movement are shown as dotted arrows in the right.

For example, a person at a given position Y can move only to the positions marked X on the right.

Without occupying any of the shaded squares at the end of each step, the minimum number of steps required to go from $\mathrm{P2}$ to $\mathrm{P5}$ is



A. 4 B. 5 C. 6 D. 7

gatecivil-2022-set1 analytical-aptitude logical-reasoning

Answer key

8.9.10 Logical Reasoning: GATE Civil 2023 Set 2 | GA Question: 6

Elvesland is a country that has peculiar beliefs and practices. They express almost all their emotions by gifting flowers. For instance, if anyone gifts a white flower to someone, then it is always taken to be a declaration of one's love for that person. In a similar manner, the gifting of a yellow flower to someone often means that one is angry with that person.

Based only on the information provided above, which one of the following sets of statement(s) can be logically inferred with *certainty*?

i. In Elvesland, one always declares one's love by gifting a white flower.
ii. In Elvesland, all emotions are declared by gifting flowers.
iii. In Elvesland, sometimes one expresses one's anger by gifting a flower that is not yellow.
iv. In Elvesland, sometimes one expresses one's love by gifting a white flower

A. only (ii) B. (i), (ii) and (iii) C. (i), (iii) and (iv) D. only (iv)

gatecivil-2023-set2 analytical-aptitude logical-reasoning

Answer key

8.9.11 Logical Reasoning: GATE ECE 2023 | GA Question: 9

Out of 1000 individuals in a town, 100 unidentified individuals are covid positive. Due to lack of adequate covid-testing kits, the health authorities of the town devised a strategy to identify these covid-positive individuals. The strategy is to:

i. Collect saliva samples from all 1000 individuals and randomly group them into sets of 5.
ii. Mix the samples within each set and test the mixed sample for covid.
iii. If the test done in (ii) gives a negative result, then declare all the 5 individuals to be covid negative.
iv. If the test done in (ii) gives a positive result, then all the 5 individuals are separately tested for covid.

Given this strategy, no more than _______ testing kits will be required to identify all the 100 covid positive individuals irrespective of how they are grouped.

A. 700 B. 600 C. 800 D. 1000

gateece-2023 analytical-aptitude logical-reasoning

Answer key

8.9.12 Logical Reasoning: GATE ECE 2024 | GA Question: 2

$\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}$, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms. All of them agree that the siblings should not share the same room.

If S and Q are single children, and the room allocations shown below are acceptable to all.

PR	TS	Q

PQ	RT	S

then, which one of the given options is the siblings?

A. P and T B. P and S
C. T and Q D. T and R

gateece-2024 logical-reasoning analytical-aptitude

Answer key

8.9.13 Logical Reasoning: GATE Electrical 2023 | GA Question: 4

A recent survey shows that 65% of tobacco users were advised to stop consuming tobacco. The survey also shows that 3 out of 10 tobacco users attempted to stop using tobacco.

Based only on the information in the above passage, which one of the following options can be logically inferred with *certainty*?

A. A majority of tobacco users who were advised to stop consuming tobacco made an attempt to do so.
B. A majority of tobacco users who were advised to stop consuming tobacco did not attempt to do so.
C. Approximately 30% of tobacco users successfully stopped consuming tobacco.
D. Approximately 65% of tobacco users successfully stopped consuming tobacco.

gate2023-ee analytical-aptitude logical-reasoning

Answer key☟

8.9.14 Logical Reasoning: GATE Electrical 2023 | GA Question: 6

Students of all the departments of a college who have successfully completed the registration process are eligible to vote in the upcoming college elections. However, by the time the due date for registration was over, it was found that suprisingly none of the students from the Department of Human Sciences had completed the registration process.

Based only on the information provided above, which one of the following sets of statement(s) can be logically inferred with *certainty*?

i. All those students who would not be eligible to vote in the college elections would certainly belong to the Department of Human Sciences.
ii. None of the students from departments other than Human Sciences failed to complete the registration process within the due time.
iii. All the eligible voters would certainly be students who are not from the Department of Human Sciences.

A. (i) and (ii)
B. (i) and (iii)
C. only (i)
D. only (iii)

gate2023-ee analytical-aptitude logical-reasoning

Answer key☟

8.9.15 Logical Reasoning: GATE IN 2023 | GA Question: 6

Residency is a famous housing complex with many well-established individuals among its residents. A recent survey conducted among the residents of the complex revealed that all of those residents who are well established in their respective fields happen to be academicians. The survey also revealed that most of these academicians are authors of some best-selling books.

Based only on the information provided above, which one of the following statements can be logically inferred with *certainty?*

A. Some residents of the complex who are well established in their fields are also authors of some best-selling books.
B. All academicians residing in the complex are well established in their fields.
C. Some authors of best-selling books are residents of the complex who are well established in their fields.
D. Some academicians residing in the complex are well established in their fields.

gatein-2023 analytical-aptitude logical-reasoning

Answer key☟

8.9.16 Logical Reasoning: GATE IN 2023 | GA Question: 8

The information contained in DNA is used to synthesize proteins that are necessary for the functioning of life. DNA is composed of four nucleotides: Adenine (A), Thymine (T), Cytosine (C), and Guanine (G). The information contained in DNA can then be thought of as a sequence of these four nucleotides: A, T, C, and G. DNA has coding and non-coding regions. Coding regions-where the sequence of these nucleotides are read in groups of three to produce individual amino acids - constitute only about 2% of human DNA. For example, the triplet of nucleotides CCG codes for the amino acid glycine, while the triplet GGA codes for the amino acid proline. Multiple amino acids are then assembled to form a protein.

Based only on the information provided above, which of the following statements can be logically inferred with certainty?

i. The majority of human DNA has no role in the synthesis of proteins.
ii. The function of about 98% of human DNA is not understood.

A. only (i)
B. only (ii)
C. both (i) and (ii)
D. neither (i) nor (ii)

gatein-2023 analytical-aptitude logical-reasoning

Answer key☟

8.9.17 Logical Reasoning: GATE Mechanical 2021 Set 1 | GA Question: 4

If $\begin{cases} \text{“}\oplus\text{” means “}-\text{”} \\ \text{“}\otimes\text{” means “}\div\text{”} \\ \text{“}\triangle\text{” means “}+\text{”} \\ \text{“}\nabla\text{” means “}\times\text{”} \end{cases}$

then, the value of the expression $\triangle 2 \oplus 3 \triangle \left((4 \otimes 2) \nabla 4 \right) =$

A. -1
B. -0.5
C. 6
D. 7

gateme-2021-set1 analytical-aptitude logical-reasoning

Answer key☟

8.9.18 Logical Reasoning: GATE Mechanical 2021 Set 2 | GA Question: 4

If $\bigoplus \div \bigodot = 2;\ \bigoplus \div \Delta = 3;\ \bigodot + \Delta = 5;\ \Delta \times \bigotimes = 10,$

Then, the value of $\left(\bigotimes - \bigoplus\right)^2$, is :

A. 0
B. 1
C. 4
D. 16

gateme-2021-set2 analytical-aptitude logical-reasoning

Answer key☟

8.9.19 Logical Reasoning: GATE Mechanical 2022 Set 1 | GA Question: 4

Four girls P, Q, R and S are studying languages in a University. P is learning French and Dutch. Q is learning Chinese and Japanese. R is learning Spanish and French. S is learning Dutch and Japanese.

Given that: French is easier than Dutch; Chinese is harder than Japanese; Dutch is easier than Japanese, and Spanish is easier than French.

Based on the above information, which girl is learning the most difficult pair of languages?

A. P
B. Q
C. R
D. S

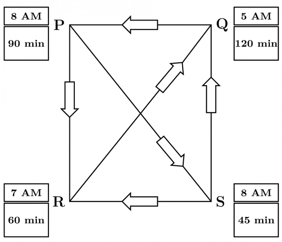
gateme-2022-set1 analytical-aptitude logical-reasoning

Answer key☟

8.9.20 Logical Reasoning: GATE Mechanical 2022 Set 2 | GA Question: 9

Four cities P, Q, R and S are connected through one-way routes as shown in the figure. The travel time between any two connected cities is one hour. The boxes beside each city name describe the starting time of first train of the day and their frequency of operation. For example, from city P, the first trains of the day start at 8 AM with a frequency of 90 minutes to each of R and S. A person does not spend additional time at any city other than the waiting time for the next connecting train.

If the person starts from R at 7 AM and is required to visit S and return to R, what is the minimum time required?



A. 6 hours 30 minutes
B. 3 hours 45 minutes
C. 4 hours 30 minutes
D. 5 hours 15 minutes

gateme-2022-set2 analytical-aptitude logical-reasoning

Answer key☟

8.9.21 Logical Reasoning: GATE2011 GG: GA-8

Three sisters $(R, S,$ and $T)$ received a total of 24 toys during Christmas. The toys were initially divided among them in a certain proportion. Subsequently, R gave some toys to S which doubled the share of S. Then S in turn gave some of her toys to T, which doubled T's share. Next, some of T's toys were given to R, which doubled the number of toys that R currently had. As a result of all such exchanges, the three sisters were left with equal number of toys. How many toys did R have originally?

A. 8
B. 9
C. 11
D. 12

gate2011-gg logical-reasoning analytical-aptitude

Answer key☟

8.9.22 Logical Reasoning: GATE2011 MN: GA-63

L, M and N are waiting in a queue meant for children to enter the zoo. There are 5 children between L and M, and 8 children between M and N. If there are 3 children ahead of N and 21 children behind L, then what is the minimum number of children in the queue?

A. 28
B. 27
C. 41
D. 40

analytical-aptitude gate2011-mn logical-reasoning

Answer key☟

8.9.23 Logical Reasoning: GATE2012 AR: GA-8

Ravi is taller than Arun but shorter than Iqbal. Sam is shorter than Ravi. Mohan is shorter than Arun. Balu is taller than Mohan and Sam. The tallest person can be

A. Mohan
B. Ravi
C. Balu
D. Arun

gate2012-ar logical-reasoning

Answer key☟

8.9.24 Logical Reasoning: GATE2012 CY: GA-9

There are eight bags of rice looking alike, seven of which have equal weight and one is slightly heavier. The weighing balance is of unlimited capacity. Using this balance, the minimum number of weighings required to identify the heavier bag is

A. 2
B. 3
C. 4
D. 8

gate2012-cy analytical-aptitude logical-reasoning

Answer key☟

8.9.25 Logical Reasoning: GATE2014 AE: GA-7

Anuj, Bhola, Chandan, Dilip, Eswar and Faisal live on different floors in a six-storeyed building (the ground floor is numbered 1, the floor above it 2, and so on) Anuj lives on an even-numbered floor, Bhola does not live on an odd numbered floor. Chandan does not live on any of the floors below Faisal's floor. Dilip does not live on floor number 2. Eswar does not live on a floor immediately above or immediately below Bhola. Faisal lives three floors above Dilip. Which of the following floor-person combinations is correct?

	Anuj	Bhola	Chandan	Dilip	Eswar	Faisal
(A)	6	2	5	1	3	4
(B)	2	6	5	1	3	4
(C)	4	2	6	3	1	5
(D)	2	4	6	1	3	5

gate2014-ae logical-reasoning analytical-aptitude descriptive

Answer key

8.9.26 Logical Reasoning: GATE2014 EC-2: GA-7

Lights of four colors (red, blue, green, yellow) are hung on a ladder. On every step of the ladder there are two lights. If one of the lights is red, the other light on that step will always be blue. If one of the lights on a step is green, the other light on that step will always be yellow. Which of the following statements is not necessarily correct?

A. The number of red lights is equal to the number of blue lights
B. The number of green lights is equal to the number of yellow lights
C. The sum of the red and green lights is equal to the sum of the yellow and blue lights
D. The sum of the red and blue lights is equal to the sum of the green and yellow lights

gate2014-ec-2 analytical-aptitude logical-reasoning normal

Answer key

8.9.27 Logical Reasoning: GATE2015 EC-1: GA-4

Operators $\square, \lozenge$ and $\rightarrow$ are defined by: $a \square b = \frac{a-b}{a+b}$; $a \lozenge b = \frac{a+b}{a-b}$; $a \rightarrow b = ab$.

Find the value of $(66 \square 6) \rightarrow (66 \lozenge 6)$.

A. -2 B. -1 C. 1 D. 2

gate2015-ec-1 logical-reasoning easy

Answer key

8.9.28 Logical Reasoning: GATE2016 EC-1: GA-5

Michael lives 10 km away from where I live. Ahmed lives 5 km away and Susan lives 7 km away from where I live. Arun is farther away than Ahmed but closer than Susan from where I live. From the information provided here, what is one possible distance (in km) at which I live from Arun's place?

A. 3.00 B. 4.99 C. 6.02 D. 7.01

gate2016-ec-1 logical-reasoning analytical-aptitude

Answer key

8.9.29 Logical Reasoning: GATE2016 EC-3: GA-8

A flat is shared by four first year undergraduate students. They agreed to allow the oldest of them to enjoy some extra space in the flat. Manu is two months older than Sravan, who is three months younger than Trideep. Pavan is one month older than Sravan. Who should occupy the extra space in the flat?

A. Manu B. Sravan C. Trideep D. Pavan

gate2016-ec-3 logical-reasoning

Answer key

8.9.30 Logical Reasoning: GATE2017 CE-1: GA-7

Students applying for hostel rooms are allotted rooms in order of seniority. Students already staying in a room will move if they get a room in their preferred list. Preferences of lower ranked applicants are ignored during allocation.

Given the data below, which room will Ajit stay in ?

Names	Student seniority	Current room	Room preference list
Amar	1	P	R, S, Q
Akbar	2	None	R, S
Anthony	3	Q	P
Ajit	4	S	Q, P, R

A. P
B. Q
C. R
D. S

gate2017-ce-1 logical-reasoning normal

Answer key☟

8.9.31 Logical Reasoning: GATE2017 CE-2: GA-3

Four cards lie on table. Each card has a number printed on one side and a colour on the other. The faces visible on the cards are $2, 3,$ red, and blue.

Proposition: If a card has an even value on one side, then its opposite face is red.

The card which MUST be turned over to verify the above proposition are

A. $2,$ red
B. $2, 3,$ red
C. $2,$ blue
D. $2,$ red, blue

gate2017-ce-2 logical-reasoning propositional-logic

Answer key☟

8.9.32 Logical Reasoning: GATE2017 EC-2: GA-3

A rule states that in order to drink beer, one must be over 18 years old. In a bar, there are 4 people. P is 16 years old. Q is 25 years old. R is drinking milkshake and S is drinking a beer. What must be checked to ensure that the rule is being followed?

A. Only P's drink
B. Only P's drink and S's age
C. Only S's age
D. Only P's drink, Q's drink and S's age

gate2017-ec-2 general-aptitude analytical-aptitude logical-reasoning

Answer key☟

8.9.33 Logical Reasoning: GATE2017 ME-1: GA-5

P, Q and R talk about $S's$ car collection. P states that S has at least 3 cars. Q believes that S has less than 3 cars. R indicates that to his knowledge, S has at least one car. Only one of P, Q and R is right. The number of cars owned by S is.

A. 0
B. 1
C. 3
D. Cannot be determined.

gate2017-me-1 general-aptitude logical-reasoning

Answer key☟

8.9.34 Logical Reasoning: GATE2017 ME-2: GA-5

P looks at Q while Q looks at R. P is married, R is not. The number of pairs of people in which a married person is looking at an unmarried person is

A. 0
B. 1
C. 2
D. Cannot be determined.

gate2017-me-2 analytical-aptitude logical-reasoning

Answer key☟

8.9.35 Logical Reasoning: GATE2018 CE-1: GA-10

Each of the letters arranged as below represents a unique integer from 1 to 9. The letters are positioned in the figure such that $(A \times B \times C), (B \times G \times E)$ and $(D \times E \times F)$ are equal. Which integer among the following choices cannot be represented by the letters A, B, C, D, E, F *or* G?

A		D
B	G	E
C		F

A. 4 B. 5 C. 6 D. 9

gate2018-ce-1 general-aptitude analytical-aptitude logical-reasoning

Answer key

8.9.36 Logical Reasoning: GATE2018 EE: GA-10

$P, Q, R,$ and S crossed a lake in a boat that can hold a maximum of two persons, with only one set of oars. The following additional facts are available.

i. The boat held two persons on each of the three forward trips across the lake and one person on each of the two return trips.
ii. P is unable to row when someone else is in the boat.
iii. Q is unable to row with anyone else except R.
iv. Each person rowed for at least one trip.
v. Only one person can row during a trip.

Who rowed twice?

A. P B. Q C. R D. S

analytical-aptitude normal gate2018-ee logical-reasoning

Answer key

8.9.37 Logical Reasoning: GATE2019 CE-2: GA-6

Mohan, the manager, wants his four workers to work in pairs. No pair should work for more than 5 hours. Ram and John have worked together for 5 hours. Krishna and Amir have worked as a team for 2 hours. Krishna does not want to work with Ram. Whom should Mohan allot to work with John, if he wants all the workers to continue working?

A. Amir B. Krishna C. Ram D. None of three

gate2019-ce-2 general-aptitude logical-reasoning

Answer key

8.9.38 Logical Reasoning: GATE2019 EE: GA-10

Consider five people- Mita, Ganga, Rekha, Lakshmi, and Sana. Ganga is taller than both Rekha and Lakshmi. Lakshmi is taller than Sana. Mita is taller than Ganga.

Which of the following conclusions are true?

1. Lakshmi is taller than Rekha
2. Rekha is shorter than Mita
3. Rekha is taller than Sana
4. Sana is shorter than Ganga

A. 1 and 3 B. 3 only C. 2 and 4 D. 1 only

gate2019-ee general-aptitude logical-reasoning

Answer key

8.9.39 Logical Reasoning: GATE2019 IN: GA-4

Five numbers $10, 7, 5, 4$ and 2 are to be arranged in a sequence from left to right following the directions given below:

1. No two odd or even numbers are next to each other.
2. The second number from the left is exactly half of the left-most number.
3. The middle number is exactly twice the right-most number.

Which is the second number from the right?

A. 2 B. 4 C. 7 D. 10

gate2019-in general-aptitude analytical-aptitude logical-reasoning

Answer key

8.10 Number Relations (3)

8.10.1 Number Relations: GATE CSE 2024 | Set 2 | GA Question: 10

In the 4×4 array shown below, each cell of the first three rows has either a cross (X) or a number.

1	X	4	3
X	5	5	4
3	X	6	X

The number in a cell represents the count of the immediate neighboring cells (left, right, top, bottom, diagonals) NOT having a cross (X). Given that the last row has no crosses (X), the sum of the four numbers to be filled in the last row is

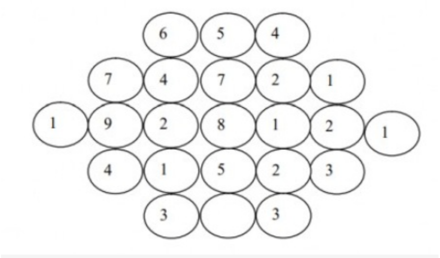
A. 11 B. 10 C. 12 D. 9

gatecse2024-set2 analytical-aptitude number-relations two-marks

Answer key

8.10.2 Number Relations: GATE2015 EC-1: GA-8

Fill in the missing value

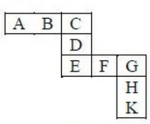


gate2015-ec-1 analytical-aptitude numerical-answers number-relations

Answer key

8.10.3 Number Relations: GATE2018 CE-2: GA-10

Each of the letters in the figure below represents a unique integer from 1 to 9. The letters are positioned in the figure such that each of $(A+B+C), (C+D+E), (E+F+G)$ and $(G+H+K)$ is equal to 13. Which integer does E represent?



A. 1 B. 4 C. 6 D. 7

gate2018-ce-2 general-aptitude analytical-aptitude number-relations

Answer key

8.11 Odd One (6)

8.11.1 Odd One: GATE CSE 2016 Set 2 | Question: GA-04

Pick the odd one from the following options.

A. CADBE B. JHKIL C. XVYWZ D. ONPMQ

gatecse-2016-set2 analytical-aptitude odd-one normal

Answer key

8.11.2 Odd One: GATE Mechanical 2024 | GA Question: 2

Find the odd one out in the set: $\{19, 37, 21, 17, 23, 29, 31, 11\}$

A. 21
B. 29
C. 37
D. 23

gateme-2024 analytical-aptitude odd-one

Answer key

8.11.3 Odd One: GATE2014 AE: GA-6

Find the odd one in the following group: $\text{ALRVX, EPVZB, ITZDF, OYEIK}$

A. ALRVX
B. EPVZB
C. ITZDF
D. OYEIK

gate2014-ae odd-one analytical-aptitude

Answer key

8.11.4 Odd One: GATE2014 EC-1: GA-6

Find the odd one from the following group:

$W,E,K,O \quad I,Q,W,A \quad F,N,T,X \quad N,V,B,D$

A. W,E,K,O
B. I,Q,W,A
C. F,N,T,X
D. N,V,B,D

gate2014-ec-1 odd-one normal

Answer key

8.11.5 Odd One: GATE2014 EC-2: GA-6

Find the odd one in the following group

$Q,W,Z,B \quad B,H,K,M \quad W,C,G,J \quad M,S,V,X$

A. Q,W,Z,B
B. B,H,K,M
C. W,C,G,J
D. M,S,V,X

gate2014-ec-2 analytical-aptitude odd-one normal

Answer key

8.11.6 Odd One: GATE2016 EC-3: GA-4

The number that least fits this set: $(324, 441, 97$ and $64)$ is ________.

A. 324
B. 441
C. 97
D. 64

gate2016-ec-3 odd-one analytical-aptitude

Answer key

8.12 Passage Reading (6)

8.12.1 Passage Reading: GATE CH 2022 | GA Question: 6

Rice, a versatile and inexpensive source of carbohydrate, is a critical component of diet worldwide. Climate change, causing extreme weather, poses a threat to sustained availability of rice. Scientists are working on developing Green Super Rice (GSR), which is resilient under extreme weather conditions yet gives higher yields sustainably.

Which one of the following is the CORRECT logical inference based on the information given in the above passage?

A. GSR is an alternative to regular rice, but it grows only in an extreme weather
B. GSR may be used in future in response to adverse effects of climate change
C. GSR grows in an extreme weather, but the quantity of produce is lesser than regular ice
D. Regular rice will continue to provide good yields even in extreme weather

gatech-2022 analytical-aptitude logical-reasoning passage-reading

Answer key☟

8.12.2 Passage Reading: GATE CSE 2022 | GA Question: 7

In a recently conducted national entrance test, boys constituted 65% of those who appeared for the test. Girls constituted the remaining candidates and they accounted for 60% of the qualified candidates.

Which one of the following is the correct logical inference based on the information provided in the above passage?

A. Equal number of boys and girls qualified
B. Equal number of boys and girls appeared for the test
C. The number of boys who appeared for the test is less than the number of girls who appeared
D. The number of boys who qualified the test is less than the number of girls who qualified

gatecse-2022 analytical-aptitude logical-reasoning passage-reading two-marks

Answer key☟

8.12.3 Passage Reading: GATE CSE 2023 | GA Question: 6

The country of Zombieland is in distress since more than 75% of its working population is suffering from serious health issues. Studies conducted by competent health experts concluded that a complete lack of physical exercise among its working population was one of the leading causes of their health issues. As one of the measures to address the problem, the Government of Zombieland has decided to provide monetary incentives to those who ride bicycles to work.

Based only on the information provided above, which one of the following statements can be logically inferred with *certainty*?

A. All the working population of Zombieland will henceforth ride bicycles to work.
B. Riding bicycles will ensure that all of the working population of Zombieland is free of health issues.
C. The health experts suggested to the Government of Zombieland to declare riding bicycles as mandatory.
D. The Government of Zombieland believes that riding bicycles is a form of physical exercise.

gatecse-2023 analytical-aptitude logical-reasoning passage-reading two-marks

Answer key☟

8.12.4 Passage Reading: GATE ECE 2022 | GA Question: 6

Mosquitoes pose a threat to human health. Controlling mosquitoes using chemicals may have undesired consequences. In Florida, authorities have used genetically modified mosquitoes to control the overall mosquito population. It remains to be seen if this novel approach has unforeseen consequences.

Which one of the following is the correct logical inference based on the information in the above passage?

A. Using chemicals to kill mosquitoes is better than using genetically modified mosquitoes because genetic engineering is dangerous
B. Using genetically modified mosquitoes is better than using chemicals to kill mosquitoes because they do not have any side effects
C. Both using genetically modified mosquitoes and chemicals have undesired consequences and can be dangerous
D. Using chemicals to kill mosquitoes may have undesired consequences but it is not clear if using genetically modified mosquitoes has any negative consequence

gateece-2022 analytical-aptitude logical-reasoning passage-reading

Answer key☟

8.12.5 Passage Reading: GATE Electrical 2020 | GA Question: 6

Non-performing Assets (NPAs) of a bank in India is defined as an asset, which remains unpaid by a borrower for a certain period of time in terms of interest, principal, or both. Reserve Bank of India (RBI) has changed the definition of NPA thrice during $1993-2004$, in terms of the holding period of loans. The holding period was reduced by one quarter each time. In 1993, the holding period was four quarters (360 days).

Based on the above paragraph, the holding period of loans in 2004 after the third revision was ______ days.

A. 45 B. 90 C. 135 D. 180

gate2020-ee analytical-aptitude logical-reasoning passage-reading

Answer key

8.12.6 Passage Reading: GATE Electrical 2022 | GA Question: 6

Altruism is the human concern for the wellbeing of others. Altruism has been shown to be motivated more by social bonding, familiarity and identification of belongingness to a group. The notion that altruism may be attributed to empathy or guilt has now been rejected.

Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Humans engage in altruism due to guilt but not empathy
B. Humans engage in altruism due to empathy but not guilt
C. Humans engage in altruism due to group identification but not empathy
D. Humans engage in altruism due to empathy but not familiarity

gate2022-ee analytical-aptitude logical-reasoning passage-reading

Answer key

8.13 Round Table Arrangement (6)

8.13.1 Round Table Arrangement: GATE CH 2022 | GA Question: 4

Six persons $\text{P, Q, R, S, T and U}$ are sitting around a circular table facing the center not necessarily in the same order. Consider the following statements:

- P sits next to S and T.
- Q sits diametrically opposite to P.
- The shortest distance between S and R is equal to the shortest distance between T and U.

Based on the above statements, Q is a neighbor of

A. U and S
B. R and T
C. R and U
D. P and S

gatech-2022 analytical-aptitude round-table-arrangement

Answer key

8.13.2 Round Table Arrangement: GATE CSE 2017 Set 1 | Question: GA-7

Six people are seated around a circular table. There are at least two men and two women. There are at least three right-handed persons. Every woman has a left-handed person to her immediate right. None of the women are right-handed. The number of women at the table is

A. 2
B. 3
C. 4
D. Cannot be determined

gatecse-2017-set1 analytical-aptitude round-table-arrangement

Answer key

8.13.3 Round Table Arrangement: GATE Chemical 2020 | GA Question: 7

$P, Q, R, S, T, U, V,$ and W are seated around a circular table.

I. S is seated opposite to W
II. U is seated at the second place to the right of R
III. T is seated at the third place to the left of R
IV. V is a neighbour of S

Which of the following must be true?

A. P is a neighbour of R
B. Q is a neighbour of R
C. P is not seated opposite to Q
D. R is the left neighbour of S

gate2020-ch analytical-aptitude logical-reasoning round-table-arrangement

Answer key

8.13.4 Round Table Arrangement: GATE Civil 2023 Set 2 | GA Question: 7

Three husband-wife pairs are to be seated at a circular table that has six identical chairs. Seating arrangements are defined only by the relative position of the people. How many seating arrangements are possible such that every husband sits next to his wife?

A. 16 B. 4 C. 120 D. 720

gatecivil-2023-set2 analytical-aptitude round-table-arrangement

Answer key☟

8.13.5 Round Table Arrangement: GATE2017 CE-2: GA-8

P, Q, R, S, T, and U are seated around a circular table . R is seated two places to the right of Q. P is seated three places to the left of R. S is seated opposite U. If P and U now switch seats, which of the following must necessarily be true?

A. P is immediately to the right of R
B. T is immediately to the left of P
C. T is immediately to the left of P or P is immediately to the right of Q
D. U is immediately to the right of R or P is immediately to the left of T

gate2017-ce-2 logical-reasoning round-table-arrangement

Answer key☟

8.13.6 Round Table Arrangement: GATE2017 EC-1: GA-7

S, T, U, V, W, X, Y and Z are seated around a circular table. $T's$ neighbors are Y and V. Z is seated third to the left of T and second to the right of S. $U's$ neighbors are S and Y; and T and W are not seated opposite each other. Who is third to the left of V?

A. X B. W C. U D. T

gate2017-ec-1 analytical-aptitude round-table-arrangement

Answer key☟

8.14 Seating Arrangement (4)

8.14.1 Seating Arrangement: GATE CSE 2017 Set 1 | Question: GA-3

Rahul, Murali, Srinivas and Arul are seated around a square table. Rahul is sitting to the left of Murali. Srinivas is sitting to the right of Arul. Which of the following pairs are seated opposite each other?

A. Rahul and Murali B. Srinivas and Arul C. Srinvas and Murali D. Srinivas and Rahul

gatecse-2017-set1 analytical-aptitude logical-reasoning seating-arrangement

Answer key☟

8.14.2 Seating Arrangement: GATE Civil 2020 Set 1 | GA Question: 7

Five friends P, Q, R, S and T went camping. At night, they had to sleep in a row inside the tent. P, Q, and T refused to sleep next to R since he snored loudly. P and S wanted to avoid Q as he usually hugged people in sleep.

Assuming everyone was satisfied with the sleeping arrangements, what is the order in which they slept?

A. $RSPTQ$ B. $SPRTQ$ C. $QRSPT$ D. $QTSPR$

gate2020-ce-1 analytical-aptitude logical-reasoning seating-arrangement

Answer key☟

8.14.3 Seating Arrangement: GATE Electrical 2021 | GA Question: 10

Seven cars $\text{P, Q, R, S, T, U and V}$ are parked in a row not necessarily in that order. The cars T and U should be parked next to each other. The cars S and V also should be parked next to each other, whereas P and Q cannot be parked next to each other. Q and S must be parked next to each other. R is parked to the immediate right of V. T is parked to the left of U.

Based on the above statements, the only INCORRECT option given below is:

A. There are two cars parked in between Q and V.
B. Q and R are not parked together.
C. V is the only car parked in between S and R.
D. Car P is parked at the extreme end.

gateee-2021 analytical-aptitude logical-reasoning seating-arrangement

Answer key☟

8.14.4 Seating Arrangement: GATE2019 EC: GA-6

Four people are standing in a line facing you. They are Rahul, Mathew, Seema and Lohit. One is an engineer, one is a doctor, one a teacher and another a dancer. You are told that:

1. Mathew is not standing next to Seema
2. There are two people standing between Lohit and the engineer
3. Rahul is not a doctor
4. The teacher and the dancer are standing next to each other
5. Seema is turning to her right to speak to the doctor standing next to her

Who among them is an engineer?

A. Seema
B. Lohit
C. Rahul
D. Mathew

gate2019-ec general-aptitude analytical-aptitude logical-reasoning seating-arrangement

Answer key☟

8.15 Sequence Series (12)

8.15.1 Sequence Series: GATE CSE 2020 | Question: GA-7

If $P=3$, $R=27$, $T=243$, then $Q+S=$ ________

A. 40
B. 80
C. 90
D. 110

gatecse-2020 analytical-aptitude logical-reasoning sequence-series two-marks

Answer key☟

8.15.2 Sequence Series: GATE CSE 2024 | Set 2 | GA Question: 5

In the sequence $6, 9, 14, x, 30, 41$, a possible value of x is

A. 25
B. 21
C. 18
D. 20

gatecse2024-set2 analytical-aptitude sequence-series one-mark

Answer key☟

8.15.3 Sequence Series: GATE Civil 2020 Set 1 | GA Question: 4

If $0, 1, 2, \ldots, 7, 8, 9$ are coded as $O, P, Q, \ldots, V, W, X$, then 45 will be coded as ______.

A. TS
B. ST
C. SS
D. SU

gate2020-ce-1 analytical-aptitude logical-reasoning sequence-series easy

Answer key☟

8.15.4 Sequence Series: GATE Civil 2024 Set 1 | GA Question: 4

Which one of the given options is a possible value of x in the following sequence?

$$3, 7, 15, x, 63, 127, 255$$

A. 35
B. 40
C. 45
D. 31

gatecivil-2024-set1 analytical-aptitude sequence-series

Answer key☟

8.15.5 Sequence Series: GATE Electrical 2020 | GA Question: 7

Select the next element of the series: Z, WV , RQP, _____

A. LKJI B. JIHG C. KJIH D. NMLK

gate2020-ee analytical-aptitude logical-reasoning sequence-series

Answer key☟

8.15.6 Sequence Series: GATE Electrical 2020 | GA Question: 8

In four-digit integer numbers from 1001 to 9999, the digit group "37" (in the same sequence) appears ________ times.

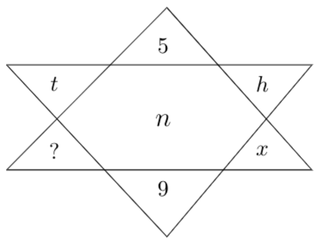
A. 270 B. 279 C. 280 D. 299

gate2020-ee analytical-aptitude logical-reasoning sequence-series

Answer key☟

8.15.7 Sequence Series: GATE Mechanical 2020 Set 2 | GA Question: 7

Find the missing element in the following figure.



A. d B. e C. w D. y

gateme-2020-set2 analytical-aptitude logical-reasoning sequence-series

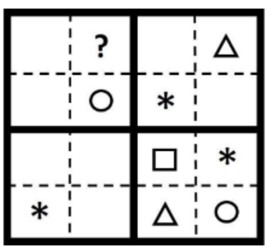
Answer key☟

8.15.8 Sequence Series: GATE Mechanical 2023 | GA Question: 5

The symbols $\bigcirc, *, \triangle$, and $\square$ are to be filled, one in each box, as shown below.
The rules for filling in the four symbols are as follows.

1. Every row and every column must contain each of the four symbols.
2. Every 2×2 square delineated by bold lines must contain each of the four symbols.

Which symbol will occupy the box marked with **'?'** in the partially filled figure?



A. $\bigcirc$ B. $*$ C. $\triangle$ D. $\square$

gateme-2023 analytical-aptitude sequence-series

Answer key☟

8.15.9 Sequence Series: GATE2010 MN: GA-7

Given the sequence $A, B, B, C, C, C, D, D, D, D, \ldots$ etc. , that is one A, two $B's$, three $C's$, four $D's$, five $E's$ and so on, the 240^{th} letter in the sequence will be :

A. V B. U C. T D. W

general-aptitude logical-reasoning gate2010-mn sequence-series

Answer key

8.15.10 Sequence Series: GATE2014 EC-3: GA-6

Find the next term in the sequence: $7G, 11K, 13M$, ________.

A. $15Q$ B. $17Q$ C. $15P$ D. $17P$

gate2014-ec-3 analytical-aptitude logical-reasoning sequence-series

Answer key

8.15.11 Sequence Series: GATE2015 EC-3: GA-4

Find the missing sequence in the letter series below:
$A, CD, GHI, ?, UVWXY$

A. LMN B. MNO C. $MNOP$ D. $NOPQ$

gate2015-ec-3 general-aptitude logical-reasoning sequence-series

Answer key

8.15.12 Sequence Series: GATE2018 ME-2: GA-3

Find the missing group of letters in the following series:
$BC, FGH, LMNO,$ ______

A. $UVWXY$ B. $TUVWX$ C. $STUVW$ D. $RSTUV$

gate2018-me-2 analytical-aptitude logical-reasoning sequence-series

Answer key

8.16 Statements Follow (33)

8.16.1 Statements Follow: GATE CH 2022 | GA Question: 9

Given below are three statements and four conclusions drawn based on the statements.

Statement 1 : Some engineers are writers.

Statement 2 : No writer is an actor.

Statement 3 : All actors are engineers.

Conclusion I : Some writers are engineers.

Conclusion II : All engineers are actors.

Conclusion III : No actor is a writer.

Conclusion IV : Some actors are writers.

Which one of the following options can be logically inferred?

A. Only conclusion I is correct
B. Only conclusion II and conclusion III are correct
C. Only conclusion I and conclusion III are correct
D. Either conclusion III or conclusion IV is correct

gatech-2022 analytical-aptitude statements-follow

Answer key

8.16.2 Statements Follow: GATE CH 2023 | GA Question: 6

Human beings are one among many creatures that inhabit an imagined world. In this imagined world, some creatures are cruel. If in this imagined world, it is given that the statement "Some human beings are not cruel creatures" is FALSE, then which of the following set of statement(s) can be logically inferred with *certainty*?

i. All human beings are cruel creatures.

ii. Some human beings are cruel creatures.
iii. Some creatures that are cruel are human beings.
iv. No human beings are cruel creatures.

A. only (i)
B. only (iii) and (iv)
C. only (i) and (ii)
D. (i), (ii) and (iii)

gatech-2023 analytical-aptitude statements-follow

Answer key

8.16.3 Statements Follow: GATE CSE 2016 Set 1 | Question: GA08

Consider the following statements relating to the level of poker play of four players $P, Q, R\ and\ S$.

I. P always beats Q
II. R always beats S
III. S loses to P only sometimes.
IV. R always loses to Q

Which of the following can be logically inferred from the above statements?

i. P is likely to beat all the three other players
ii. S is the absolute worst player in the set

A. (i). only
B. (ii) only
C. (i) and (ii) only'
D. neither (i) nor (ii)

gatecse-2016-set1 analytical-aptitude normal statements-follow

Answer key

8.16.4 Statements Follow: GATE CSE 2016 Set 2 | Question: GA-08

All hill-stations have a lake. Ooty has two lakes.

Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

i. Ooty is not a hill-station.
ii. No hill-station can have more than one lake.

A. (i) only.
B. (ii) only.
C. Both (i) and (ii)
D. Neither (i) nor (ii)

gatecse-2016-set2 analytical-aptitude easy statements-follow

Answer key

8.16.5 Statements Follow: GATE CSE 2021 Set 1 | GA Question: 9

Given below are two statements 1 and 2, and two conclusions I and II

- Statement 1: All bacteria are microorganisms.
- Statement 2: All pathogens are microorganisms.

- Conclusion I: Some pathogens are bacteria.
- Conclusion II: All pathogens are not bacteria.

Based on the above statements and conclusions, which one of the following options is logically CORRECT?

A. Only conclusion I is correct
B. Only conclusion II is correct
C. Either conclusion I or II is correct
D. Neither conclusion I nor II is correct

gatecse-2021-set1 analytical-aptitude logical-reasoning statements-follow two-marks

Answer key

8.16.6 Statements Follow: GATE CSE 2022 | GA Question: 4

Given below are four statements.

- Statement 1 : All students are inquisitive.
- Statement 2 : Some students are inquisitive.
- Statement 3 : No student is inquisitive.
- Statement 4 : Some students are not inquisitive.

From the given four statements, find the two statements that CANNOT BE TRUE simultaneously, assuming that there is at least one student in the class.

A. Statement 1 and Statement 3
B. Statement 1 and Statement 2
C. Statement 2 and Statement 4
D. Statement 3 and Statement 4

gatecse-2022 analytical-aptitude logical-reasoning statements-follow one-mark

Answer key☟

8.16.7 Statements Follow: GATE Civil 2021 Set 2 | GA Question: 8

1. Some football players play cricket.
2. All cricket players play hockey.

Among the options given below, the statement that logically follows from the two statements 1 and 2 above, is :

A. No football player plays hockey
B. Some football players play hockey
C. All football players play hockey
D. All hockey players play football

gatecivil-2021-set2 analytical-aptitude logical-reasoning statements-follow

Answer key☟

8.16.8 Statements Follow: GATE Civil 2022 Set 2 | GA Question: 9

Given below are two statements and four conclusions drawn based on the statements.

- Statement 1 : Some soaps are clean.
- Statement 2 : All clean objects are wet.

Conclusion I: Some clean objects are soaps.

Conclusion II: No clean object is a soap.

Conclusion III: Some wet objects are soaps.

Conclusion IV: All wet objects are soaps.

Which one of the following options can be logically inferred?

A. Only conclusion I is correct
B. Either conclusion I or conclusion II is correct
C. Either conclusion III or conclusion IV is correct
D. Only conclusion I and conclusion III are correct

gatecivil-2022-set2 analytical-aptitude statements-follow

Answer key☟

8.16.9 Statements Follow: GATE Civil 2023 Set 1 | GA Question: 4

A duck named Donald Duck says "All ducks always lie."

Based only on the information above, which one of the following statements can be logically inferred with *certainty*?

A. Donald Duck always lies.
B. Donald Duck always tells the truth.
C. Donald Duck's statement is true.
D. Donald Duck's statement is false.

gatecivil-2023-set1 analytical-aptitude statements-follow

Answer key☟

8.16.10 Statements Follow: GATE Civil 2023 Set 1 | GA Question: 6

Based only on the truth of the statement 'Some humans are intelligent', which one of the following options can be logically inferred with *certainty*?

A. No human is intelligent.
B. All humans are intelligent.
C. Some non-humans are intelligent.
D. Some intelligent beings are humans.

gatecivil-2023-set1 analytical-aptitude statements-follow

Answer key☟

8.16.11 Statements Follow: GATE Civil 2024 Set 2 | GA Question: 2

Statements:

1. All heroes are winners.
2. All winners are lucky people.

Inferences:

I. All lucky people are heroes.
II. Some lucky people are heroes.
III. Some winners are heroes.

Which of the above inferences can be logically deduced from statements 1 and 2?

A. Only I and II B. Only II and III C. Only I and III D. Only III

gatecivil-2024-set2 analytical-aptitude logical-reasoning statements-follow

Answer key☟

8.16.12 Statements Follow: GATE DS&AI 2024 | GA Question: 6

Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk.

Based on the given passage, which of the following can be inferred?

A. All human beings can digest dairy milk.
B. No human being can digest dairy milk.
C. Digestion of dairy milk is essential for human beings.
D. In human beings, digestion of dairy milk resulted from a mutated gene.

gate-ds-ai-2024 analytical-aptitude logical-reasoning statements-follow two-marks

Answer key☟

8.16.13 Statements Follow: GATE ECE 2021 | GA Question: 6

Given below are two statements and two conclusions.

- Statement 1: All purple are green.
- Statement 2: All black are green.

- Conclusion I: Some black are purple.
- Conclusion II: No black is purple.

Based on the above statements and conclusions, which one of the following options is logically CORRECT?

A. Only conclusion I is correct
B. Only conclusion II is correct
C. Either conclusion I or II is correct
D. Both conclusion I and II are correct

gateec-2021 analytical-aptitude logical-reasoning statements-follow

Answer key☟

8.16.14 Statements Follow: GATE ECE 2022 | GA Question: 9

In a class of five students P, Q, R, S and T, only one student is known to have copied in the exam. The disciplinary committee has investigated the situation and recorded the statements from the students as given below.

Statement of P: R has copied in the exam.

Statement of Q: S has copied in the exam.

Statement of R: P did not copy in the exam.

Statement of S: Only one of us is telling the truth.

Statement of T: R is telling the truth.

The investigating team had authentic information that S never lies.

Based on the information given above, the person who has copied in the exam is

A. R B. P C. Q D. T

gateece-2022 analytical-aptitude logical-reasoning statements-follow

Answer key

8.16.15 Statements Follow: GATE ECE 2023 | GA Question: 6

Forestland is a planet inhabited by different kinds of creatures. Among other creatures, it is populated by animals all of whom are ferocious. There are also creatures that have claws and some that do not. All creatures that have claws are ferocious.

Based only on the information provided above, which one of the following options can be logically inferred with *certainty?*

A. All creatures with claws are animals.
B. Some creatures with claws are non-ferocious.
C. Some non-ferocious creatures have claws.
D. Some ferocious creatures are creatures with claws.

gateece-2023 analytical-aptitude logical-reasoning statements-follow

Answer key

8.16.16 Statements Follow: GATE Electrical 2022 | GA Question: 4

Given below are two statements and four conclusions drawn based on the statements.

Statement 1: Some bottles are cups.

Statement 2: All cups are knives.

Conclusion I : Some bottles are knives.

Conclusion II : Some knives are cups.

Conclusion III : All cups are bottles.

Conclusion IV : All knives are cups.

Which one of the following options can be logically inferred?

A. Only conclusion I and conclusion II are correct
B. Only conclusion II and conclusion III are correct
C. Only conclusion II and conclusion IV are correct
D. Only conclusion III and conclusion IV are correct

gate2022-ee analytical-aptitude statements-follow

Answer key

8.16.17 Statements Follow: GATE Electrical 2022 | GA Question: 9

The letters P, Q, R, S, T and U are to be placed one per vertex on a regular convex hexagon, but not necessarily in the same order.

Consider the following statements:

- The line segment joining R and S is longer than the line segment joining P and Q.
- The line segment joining R and S is perpendicular to the line segment joining P and Q.
- The line segment joining R and U is parallel to the line segment joining T and Q.

Based on the above statements, which one of the following options is CORRECT?

A. The line segment joining R and T is parallel to the line segment joining Q and S
B. The line segment joining T and Q is parallel to the line joining P and U
C. The line segment joining R and P is perpendicular to the line segment joining U and Q
D. The line segment joining Q and S is perpendicular to the line segment joining R and P

gate2022-ee analytical-aptitude logical-reasoning statements-follow

Answer key

8.16.18 Statements Follow: GATE Mechanical 2021 Set 2 | GA Question: 6

Given below are two statements 1 and 2, and two conclusions I and II.

- Statement 1 : All entrepreneurs are wealthy.
- Statement 2 : All wealthy are risk seekers.

- Conclusion I : All risk seekers are wealthy.
- Conclusion II : Only some entrepreneurs are risk seekers.

Based on the above statements and conclusions, which one of the following options is CORRECT?

A. Only conclusion I is correct
B. Only conclusion II is correct
C. Neither conclusion I nor II is correct
D. Both conclusions I and II are correct

gateme-2021-set2 analytical-aptitude logical-reasoning statements-follow

Answer key

8.16.19 Statements Follow: GATE Mechanical 2022 Set 1 | GA Question: 9

Given below are three conclusions drawn based on the following three statements.

Statement 1 : All teachers are professors.
Statement 2 : No professor is a male.
Statement 3 : Some males are engineers.

Conclusion I: No engineer is a professor.
Conclusion II: Some engineers are professors.
Conclusion III: No male is a teacher.

Which one of the following options can be logically inferred?

A. Only conclusion III is correct
B. Only conclusion I and conclusion II are correct
C. Only conclusion II and conclusion III are correct
D. Only conclusion I and conclusion III are correct

gateme-2022-set1 analytical-aptitude statements-follow

Answer key

8.16.20 Statements Follow: GATE2013 AE: GA-9

- All professors are researchers
- Some scientists are professors

Which of the given conclusions is logically valid and is inferred from the above arguments?

A. All scientists are researchers
B. All professors are scientists
C. Some researchers are scientists
D. No conclusion follows

gate2013-ae analytical-aptitude logical-reasoning statements-follow

Answer key

8.16.21 Statements Follow: GATE2014 AG: GA-6

In a group of four children, Som is younger to Riaz. Shiv is elder to Ansu. Ansu is youngest in the group. Which of the following statements is/are required to find the eldest child in the group?

Statements

1. Shiv is younger to Riaz.
2. Shiv is elder to Som.

A. Statement 1 by itself determines the eldest child.
B. Statement 2 by itself determines the eldest child.
C. Statements 1 and 2 are both required to determine the eldest child.
D. Statements 1 and 2 are not sufficient to determine the eldest child.

gate2014-ag analytical-aptitude normal statements-follow

Answer key

8.16.22 Statements Follow: GATE2014 EC-1: GA-2

Read the statements:

- All women are entrepreneurs.
- Some women are doctors.

Which of the following conclusions can be logically inferred from the above statements?

A. All women are doctors
B. All doctors are entrepreneurs
C. All entrepreneurs are women
D. Some entrepreneurs are doctors

gate2014-ec-1 analytical-aptitude logical-reasoning statements-follow easy

Answer key

8.16.23 Statements Follow: GATE2015 EC-1: GA-10

Humpty Dumpty sits on a wall every day while having lunch. The wall sometimes breaks. A person sitting on the wall falls if the wall breaks.

Which one of the statements below is logically valid and can be inferred from the above sentences?

A. Humpty Dumpty always falls while having lunch
B. Humpty Dumpty does not fall sometimes while having lunch
C. Humpty Dumpty never falls during dinner
D. When Humpty Dumpty does not sit on the wall, the wall does not break

gate2015-ec-1 general-aptitude analytical-aptitude statements-follow

Answer key

8.16.24 Statements Follow: GATE2015 EC-2: GA- 7

Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows.

Statements:

I. All film stars are playback singers.
II. All film directors are film stars.

Conclusions:

I. All film directors are playback singers.
II. Some film stars are film directors.

A. Only conclusion I follows.
B. Only conclusion II follows.
C. Neither conclusion I nor II follows.
D. Both conclusions I and II follow.

gate2015-ec-2 analytical-aptitude logical-reasoning statements-follow

Answer key

8.16.25 Statements Follow: GATE2015 ME-3: GA-4

1. Tanya is older than Eric.
2. Cliff is older than Tanya.
3. Eric is older than Cliff.

If the first two statements are true, then the third statement is:

A. True
B. False
C. Uncertain
D. Data insufficient

gate2015-me-3 analytical-aptitude statements-follow

Answer key

8.16.26 Statements Follow: GATE2015 ME-3: GA-7

Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows.

Statements:

I. No manager is a leader.
II. All leaders are executives.

Conclusions:

I. No manager is an executive.
II. No executive is a manager.

A. Only conclusion I follows.
B. Only conclusion II follows.
C. Neither conclusion I nor II follows.
D. Both conclusions I and II follow.

gate2015-me-3 analytical-aptitude statements-follow

Answer key

8.16.27 Statements Follow: GATE2016 CE-2: GA-8

Fact 1: Humans are mammals.

Fact 2: Some humans are engineers.

Fact 3: Engineers build houses.

If the above statements are facts, which of the following can be logically inferred?

I. All mammals build houses.

II. Engineers are mammals.

III. Some humans are not engineers.

A. II only.
B. III only.
C. I, II and III.
D. I only.

gate2016-ce-2 analytical-aptitude logical-reasoning statements-follow

Answer key

8.16.28 Statements Follow: GATE2016 ME-2: GA-4

Fact: If it rains, then the field is wet.

Read the following statements:

i. It rains
ii. The field is not wet
iii. The field is wet
iv. It did not rain

Which one of the options given below is **NOT** logically possible, based on the given fact?

A. If (iii), then (iv).
B. If (i), then (iii).
C. If (i), then (ii).
D. If (ii), then (iv).

gate2016-me-2 analytical-aptitude statements-follow

Answer key

8.16.29 Statements Follow: GATE2017 CE-1: GA-3

Consider the following sentences:

All benches are beds. No bed is bulb. Some bulbs are lamps.

Which of the following can be inferred?

i. Some beds are lamps.
ii. Some lamps are beds.

A. Only i
B. Only ii
C. Both i and ii
D. Neither i nor ii

gate2017-ce-1 general-aptitude analytical-aptitude statements-follow

Answer key☟

8.16.30 Statements Follow: GATE2017 EC-1: GA-5

Some tables are shelves. Some shelves are chairs. All chairs are benches. Which of the following conclusion can be deduced from the preceding sentences?

i. At least one bench is a table
ii. At least one shelf is a bench
iii. At least one chair is a table
iv. All benches are chairs

A. Only i
B. Only ii
C. Only ii and iii
D. Only iv

gate2017-ec-1 general-aptitude analytical-aptitude statements-follow

Answer key☟

8.16.31 Statements Follow: GATE2017 ME-2: GA-9

All people in a certain island are either 'Knights' or 'Knaves' and each person knows every other person's identity. Knights never lie, and Knaves ALWAYS lie.

P says "Both of us are Knights". Q says "None of us are Knaves".

Which one of the following can be logically inferred from the above?

A. Both P and Q are knights.
B. P is a knight; Q is a Knave.
C. Both P and Q are Knaves.
D. The identities of P, Q cannot be determined.

gate2017-me-2 analytical-aptitude logical-reasoning statements-follow

Answer key☟

8.16.32 Statements Follow: GATE2018 ME-1: GA-10

Consider the following three statements:

i. Some roses are red.
ii. All red flowers fade quickly.
iii. Some roses fade quickly.

Which of the following statements can be logically inferred from the above statements?

A. If (i) is true and (ii) is false, then (iii) is false.
B. If (i) is true and (ii) is false, then (iii) is true.
C. If (i) and (ii) are true, then (iii) is true.
D. If (i) and (ii) are false, then (iii) is false.

gate2018-me-1 general-aptitude analytical-aptitude statements-follow

Answer key☟

8.16.33 Statements Follow: GATE2019 IN: GA-2

Some students were not involved in the strike.

If the above statement is true, which of the following conclusions is/are logically necessary?

1. Some who were involved in the strike were students.
2. No student was involved in the strike.
3. At least one student was involved in the strike.
4. Some who were not involved in the strike were students.

A. 1 and 2
B. 3
C. 4
D. 2 and 3

gate2019-in general-aptitude analytical-aptitude statements-follow

Answer key☟

Answer Keys

8.1.1	B
8.2.4	C
8.5.2	A
8.5.7	D
8.5.12	A
8.6.5	D
8.9.1	A
8.9.6	C
8.9.11	A
8.9.16	D
8.9.21	C
8.9.26	X
8.9.31	C
8.9.36	C
8.10.2	3
8.11.4	D
8.12.3	D
8.13.2	A
8.14.1	C
8.15.2	B
8.15.7	A
8.15.12	B
8.16.5	C;D
8.16.10	D
8.16.15	D
8.16.20	C
8.16.25	B
8.16.30	B

8.1.2	D
8.2.5	C
8.5.3	D
8.5.8	D
8.6.1	A
8.6.6	B
8.9.2	C
8.9.7	A
8.9.12	A
8.9.17	D
8.9.22	A
8.9.27	C
8.9.32	B
8.9.37	B
8.10.3	B
8.11.5	C
8.12.4	D
8.13.3	C
8.14.2	A
8.15.3	B
8.15.8	B
8.16.1	C
8.16.6	A
8.16.11	B
8.16.16	A
8.16.21	A
8.16.26	C
8.16.31	D

8.2.1	B
8.3.1	B
8.5.4	C
8.5.9	C
8.6.2	B
8.6.7	D
8.9.3	C
8.9.8	C
8.9.13	B
8.9.18	B
8.9.23	C
8.9.28	C
8.9.33	A
8.9.38	C
8.11.1	D
8.11.6	C;D
8.12.5	B
8.13.4	A
8.14.3	A
8.15.4	D
8.15.9	A
8.16.2	D
8.16.7	B
8.16.12	D
8.16.17	A
8.16.22	D
8.16.27	B
8.16.32	C

8.2.2	C
8.4.1	D
8.5.5	C
8.5.10	A
8.6.3	B
8.7.1	D
8.9.4	B
8.9.9	B
8.9.14	D
8.9.19	B
8.9.24	A
8.9.29	C
8.9.34	B
8.9.39	C
8.11.2	A
8.12.1	B
8.12.6	C
8.13.5	C
8.14.4	D
8.15.5	C
8.15.10	B
8.16.3	D
8.16.8	D
8.16.13	C
8.16.18	C
8.16.23	B
8.16.28	C
8.16.33	C

8.2.3	B
8.5.1	A
8.5.6	C
8.5.11	C
8.6.4	B
8.8.1	B
8.9.5	B
8.9.10	D
8.9.15	X
8.9.20	A
8.9.25	B
8.9.30	B
8.9.35	B
8.10.1	A
8.11.3	D
8.12.2	D
8.13.1	C
8.13.6	A
8.15.1	C
8.15.6	C
8.15.11	C
8.16.4	D
8.16.9	D
8.16.14	B
8.16.19	A
8.16.24	D
8.16.29	D